THE PENNSYLVANIA STATE UNIVERSITY
MATH 230 – CALCULUS AND VECTOR ANALYSIS
OFFICIAL DEPARTMENT SYLLABUS
ACADEMIC YEAR 2026–2027

This document serves as the official course policies of MATH 230. Any policies not listed in this syllabus are not considered official course policies.

Course Description

Three-dimensional analytic geometry; vectors in space; partial differentiation; double and triple integrals; integral vector calculus. Students who have passed either MATH 231 or MATH 232 may not schedule MATH 230 or MATH 230H for credit.

Enforced Prerequisite

Enforced Prerequisite at Enrollment: MATH 141 or MATH 141B or MATH 141E or MATH 141G or MATH 141H.

Required Materials

Calculus: Early Transcendentals by William Briggs, Lyle Cochran, Bernard Gillett, and Eric Schulz, 3rd ed, along with access to Pearson MyLab Math. This means you will need to purchase the online system that accompanies the textbook. You do not need a hard copy of the textbook as the eText comes with purchase of Pearson MyLab. You should check first if you have a still-active version of Pearson MyLab from when you took MATH 140/141 as we typically instruct you to purchase the multiple-semester access. There are some options for purchasing access:

- 18-Week Access: approximately $85
- 24-Month Access: approximately $145

Pearson MyLab Instructions

Students should register their Pearson MyLab account using their PSU email address with the same name that is in LionPATH and Canvas. The easiest way to sign up for MyLab is by accessing Pearson from within Canvas.

Class Attendance and Absence

<u>Regular class attendance does correlate with positive learning outcomes</u>. Instructors may decide to implement activities requiring student attendance in order to earn credit. Students who are unable to attend class or complete graded assignments for more than one week at a time will be required to present documentation to verify their significant, prolonged illness. Additionally, if a student requests makeups more than two times for missed assignments, the instructor has the right to deny such requests unless a student has a University-approved reason and/or can provide the proper documentation justifying the absence and, consequently, the missed assignment.

From UHS Policies & Patient Resources: "For routine illness-related absences, students should correspond directly with the faculty as soon as possible regarding their situation, ideally, before they miss a class, exam, or other evaluative activity. University Health Services may provide verification of illness forms for significant prolonged illnesses or injuries lasting at least a week resulting in absence from classes. When appropriate, students may request the verification during their clinician visit or send a secure message to their clinician or the Advice Nurse through myUHS at studentaffairs.psu.edu/health/myuhs. If a student wants a verification of illness from University Health Services and has received care from an outside provider for a significant, prolonged illness, they must provide appropriate documentation to the University Health Services director, 502A Student Health Center, 814-865-6555."

Email Policy

Instructors have up to 48 hours to respond to a student email. Students should be aware that time sensitive emails may not receive a response immediately. This excludes weekends and holidays. Emails sent on Friday may not receive a reply until the following week. Do not ask extensive homework questions via email. Such questions should be reserved for office hours.

Calculator Policy

A calculator, online problem solver, downloadable app, etc. is an excellent tool for crunching numbers – especially when working real-world examples with messy constants – as well as a useful study resource. If you do not have a scientific calculator, websites such as GeoGebra and Wolfram Alpha are an excellent resource for computations. Note, however, that these tools are not needed to understand and apply the concepts of the course to examples with simple coefficients (all arithmetic will be simple enough to do by hand.)

No form of calculators or electronic problem solver will be permitted on any of the quizzes or exams for this course unless otherwise specified by your instructor. Their use is a violation of Penn State's Academic Integrity Policy. An Academic Integrity Report will be filed for any unwarranted use of such a device.

Tutors and Penn State Learning

Free mathematics tutoring is available at Penn State Learning. You can find information about their online and in-person options at the following site: pennstatelearning.psu.edu/tutoring/mathematics

Office Hours

Office hours are a useful resource in any course. If you have questions about course content or progress in the course, please do not hesitate to attend office hours. Office hours are **not**, however, a time and place for instructors to re-lecture content. If you miss class, it is your responsibility to get notes from a classmate. Come prepared to office hours with questions that you may have. Additionally, office hours are time to seek clarification, but they are not a time to simply attend and get answers. Instructors do not have an obligation to give you answers to homework questions, but rather, they have an obligation to elaborate on concepts and help you to better understand the material.

Online Homework

Pearson MyLab homework will be **due each week on Tuesdays at 11:59 PM**. Please note that if you do not begin the Pearson MyLab HW, the system will not let you into the assignment to see it after the due date has passed. There will be only one dropped homework assignment from Pearson MyLab, so it is important that you work to complete them all. If you miss a Pearson MyLab due date, you can submit it late, but with a 10% penalty each day on problems completed after the due date.

Written Homework

Written homework will generally be **due each week on Thursdays at 11:59 PM**, with the exception of exam weeks (see course schedule). **No assignments will be accepted after the due date**. During the submission process, you will need to specify which problems are present on which pages (including all pages which have work, not just the first page); failure to do so will result in a zero on those problems. **All written homework must be hand-written**, otherwise it will be assigned a zero. Please see the Gradescope information on Canvas. No written homework assignments are dropped. There will be one additional homework assignment at the end of the term that will be "optional" and can be used to replace your lowest homework grade of the term. There are no extensions on Written Homework. Also, when submitting to Gradescope, pages must be properly matched to the questions on the assignment, otherwise the assignment will receive a zero.

Learning Assistant Problem Sessions (LAPS)

We have Mathematics Learning Assistants (LAs) assigned to help us in MATH 230. Each week, there will be a series of evening problem sessions scheduled in classrooms on campus that you can attend to get help and additional practice. Each evening session will focus on the content during the previous and/or current week of the term.

Learning Assistant Problem Sessions (LAPS) Attendance Incentive

As an incentive for attending LA evening sessions, you will be able to have up to three of your lowest quiz or worksheet scores replaced by attending and meeting the required number for each of the time periods listed below. **You can attend <u>up to two sessions per week that will count</u> towards the incentive**. Please note that to earn credit, you must attend the sessions for an extended period of time. In addition, you **MUST** work on the worksheets provided. This is not a time to work on homework or in-class work. The Learning Assistants can help clarify concepts but they **DO NOT** give answers. **If you are not working on the designated materials, you will <u>not</u> receive credit for the session.**

- Participate in **6 evening sessions** during weeks 2–8 = One Perfect Score Replacement
- Participate in **5 evening sessions** during weeks 9–15 = Second Perfect Score Replacement
- Participate in a **<u>combined</u> 12 LA sessions across the entire term** = Another Perfect Score Replacement

To properly receive credit for LA sessions, you must complete the attendance form. If an attendance, for whatever reason, does not get counted, the student will be asked to produce the survey confirmations as proof of attendance. Otherwise, the recorded attendance count given by the LA Program will be the official record.

Quizzes

There will be regular quizzes given during class every week (except in the first week and on exam weeks).

Quiz Makeups

- If a makeup quiz is needed, a university-approved reason is required with any proper documentation.
- No more than one makeup without penalty can be approved for a non-university absence. After this, any non-university-approved makeup request is at the discretion of the instructor. The makeup request will either be denied and a zero assigned, or a penalty will be imposed.
- Makeup quizzes must be completed by Tuesday of the following week. **NO EXCEPTIONS.**

Exams

EXAM	**DATE**	**TIME**
Exam 1	9/24 (Th)	6:15–7:30 PM
Exam 2	10/21 (W)	6:15–7:30 PM
Exam 3	11/18 (W)	6:15–7:30 PM
Comprehensive Final Exam	TBA	TBA

Rooms for examinations will be announced by your instructor at a later date and will be able to be found on the math department website. Be sure to bring your physical ID, several pencils, and an eraser to all exams.

Makeup Exams

Students who miss or cannot take a midterm examination due to a valid and documented reason, such as illness, may be allowed to take the corresponding makeup examination. The scheduled makeup exam dates are listed below.

MU EXAM	**DATE**	**TIME**
MU Exam 1	9/28 (M)	6:15–7:30 PM
MU Exam 2	10/22 (Th)	6:15–7:30 PM
MU Exam 3	11/19 (Th)	6:15–7:30 PM

Makeup Exam Policies

The Math Department offers a makeup exam for students who are unable to take the Regular midterm exam. The Makeup exam will typically be scheduled during the evening, on a different day than the Regular exam.

- Makeup exams are offered only once and will not be rescheduled.
- Students who have already completed the Regular exam are not permitted to take the Makeup exam. Doing so will result in a score of zero and may constitute an Academic Integrity violation.
- Students who miss both the Regular exam and the Makeup exam without an approved excuse will receive a score of zero.

To be eligible for a makeup exam, students must:

- Contact their instructor to inform the instructor that they are signing up for the Makeup exam and provide them with the reason for the request.
- Submit a makeup exam request using the department's online form by the published deadline in the course syllabus.
- Provide any documentation supporting their request.

Who May Take the Makeup Exam?

Students may take a makeup exam under the following circumstances:

- **University-approved absences.** Examples include, but are not limited to: official class conflicts, evening exams for other courses, religious observances that occur on the date and time of the scheduled exam, and required travel for a university-sponsored activity. Students should be prepared to provide documentation. However, providing documentation does not guarantee approval.
- **Non-university-approved absences.** At most, students may be permitted one makeup exam without penalty during the semester for a non-university-approved absence. Additional makeup requests without a university-approved reason may be permitted, but will receive a penalty.

How and When to Request a Makeup Exam

Students must submit a makeup exam request through the Math Department's online request form. The student must include the reason for needing a makeup exam and provide any required documentation. Students should assume that if approved, they will be scheduled for the Makeup exam that is listed in their course syllabus. If the student also has a conflict with the date and time of the Makeup exam, the student should include that information in their initial makeup exam request.

Deadlines

- Requests submitted by 11:59 PM on the day before the Regular exam will be reviewed before the Regular exam begins.
- Requests submitted from 12:00 AM until 6:15 PM on the day of the Regular exam may not be reviewed before the Regular exam begins.

Request Decisions

After the request has been reviewed, both the student and the instructor will receive an email with one of the following decisions:

- **APPROVED** – The email will include the date, time, and location of the Makeup exam. If the assigned Makeup time conflicts with another academic obligation, the department will assign an alternate time. Again, the student should include any information about conflicts with the Makeup exam in the initial makeup exam request.
- **PENDING ADDITIONAL DOCUMENTATION** – Additional documentation is required before a decision can be made. The student should assume the current request is denied and should resubmit their request with the requested documentation.
- **DENIED** – The student may still take the Makeup exam, but a penalty will be applied.

Students who have questions about the policy, wish to appeal a decision, or are unable to submit a request before 6:15 PM on the date of the Regular exam should contact the course coordinator. Contact information for course coordinators can be found on the Math Department website as well as in the master course syllabus for each course.

Student Instructions for Exams

- Prior to the start of any exam, students must remove all headphones, earbuds, any type of watch (including non-smartwatches), and any other electronic devices that they have on/with them.
- Students are only permitted to use a calculator if one is approved for the course they are enrolled in. Only the approved calculator for this course is permitted. **NO CALCULATORS ARE PERMITTED ON MATH 230 EXAMS OR QUIZZES.**
- All items must be placed in a backpack or bag. Students without a backpack or bag must surrender these items to the exam proctors, who will keep the items at the front of the exam room. Phones may NOT remain in any pockets.
- Students should place backpacks and bags in the row in front of their seat. To allow space for bags, students should avoid sitting in the front row of the room or seat section.
- All students are expected to bring a physical photo ID (Penn State ID, driver's license, passport, or other government-issued photo identification) for identity verification.
- Students may not have scratch paper, cheat sheets, note cards, or formula sheets unless explicitly permitted for their exam. If formula sheets are given, students must hand them in with their exam. No documents or written information from the exam may leave the exam room with the student.

Policy on Reviewing Exams and Quizzes

Questions regarding the grading of exams or missing exam grades should be directed to your section instructor as soon as possible. Students and instructors will have until the next exam takes place to review their exam or report a missing grade. Students will have one week from the date of the final exam to contact their instructor regarding questions about the final exam or a missing final exam grade. Final exams are not released. If a student wishes to review their work on the final exam, they must make an appointment with their instructor for the following academic term.
This policy also applies to quizzes – any concerns regarding quizzes and/or quiz grading need to be addressed prior to the next quiz.

Final Examination

The final examination may be scheduled on any day during the final examination period, December 14–18, 2026. Students may access their final exam schedules through LionPATH in September. Notification of conflicts is given on the student's final exam schedule. **It is the <u>student's responsibility</u> to apply for any conflict final exams during this time period.**
There are two types of conflict final examinations: direct and overload. Direct conflicts are two examinations scheduled at the same time. Overload examinations are defined as three or more examinations scheduled in consecutive time periods or within one calendar day. Students may elect to take the three or more examinations on the same day if they wish or request a conflict final examination through the university registrar.

A student must take action to request a conflict exam through LionPATH between Monday, September 28, 2026 and Sunday, October 25, 2026. Conflict final examinations cannot be scheduled through the Mathematics department.

Students who miss or cannot take the final examination due to a valid and documented reason, such as illness, may be allowed to take a makeup final examination at the beginning of the next semester.

Personal business, such as travel, employment, weddings, graduations, or attendance at public events such as concerts and sporting events are not valid excuses. Early flights home, bus tickets to leave town, and family vacations are NOT valid excuses to miss or reschedule a final exam.

Students should make plans to leave campus AFTER all their scheduled exams are completed. It is best not to book flights that leave during finals week. Instructors must give the final exams according to the University schedule and cannot give makeups or reschedules for non-valid or non-approved excuses. Forgetting the date or time of an examination is not a valid excuse. **If the student does not have a valid reason, the final exam will be a zero.** All such makeup examinations must be arranged through the instructor with the approval of the course coordinator, and students in such a situation should contact their instructors within 24 hours of the scheduled final examination. Students who have taken the original final examination are not permitted to take a makeup examination.

Grades and Grading Policy

The final grade for the course will be determined according to the following distribution:

Pearson MyLab Homework	12.5%
Written Homework	12.5%
Quizzes	15%
Exams 1–3	35%
Final Exam	25%
Total	**100%**

Letter grades for this course are set according to the following cutoffs:

Letter Grade	Score Range
A	92.50%–100%
A-	89.50%–92.49%
B+	86.50%–89.49%
B	82.50%–86.49%
B-	79.50%–82.49%
C+	76.50%–79.49%
C	69.50%–76.49%
D	59.50%–69.49%
F	$\leq$59.49%

Deferred Grades (DF)

A student who is passing a course but, because of illness or emergency, needs to complete the course material at a later time can opt for a deferred grade ('DF'). The completion of course materials must be done within the first couple of weeks following the end of the term. Please note that deferred grades are only provided when the circumstances are extenuating that prevent a student from taking a final exam.

Late-Drop

Students may add/drop a course without academic penalty within the first six calendar days of the semester. A student may late drop a course within the first twelve weeks of the semester. After the first six days and before deciding to late drop this course, each student should consult with his or her academic advisor. The late drop deadline for this semester is Friday, November 13, 2026 at 11:59 PM.

Academic Integrity

According to Penn State policy G-9: Academic Integrity, an academic integrity violation is "an intentional, unintentional, or attempted violation of course or assessment policies to gain an academic advantage or to advantage or disadvantage another student academically." Unless your instructor tells you otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and may not assist other students with papers, quizzes, exams, or other assessments. If your instructor allows you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source. You may not submit false or fabricated information, use the same academic work for credit in multiple courses, or share instructional content. Students with questions about academic integrity should ask their instructor before submitting work. Students facing allegations of academic misconduct may not drop/withdraw from the affected course unless they are cleared of wrongdoing (see G-9: Academic Integrity). Attempted drops will be prevented or reversed, and students will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Dean's List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

When searching for help with your course, your instructor will always be the best resource for advice. The University is aware of fraudulent companies and scammers who prey on a student's desire for assistance with coursework. A student who hires someone to do work on their behalf demonstrates a clear example of a violation of academic integrity highlighted above. Sharing login credentials with others is also in violation of Penn State Policy AD95 and may be subject to additional disciplinary action. Furthermore, a fraudulent company and/or scammer may extort the student by threats to communicate the student's arrangement with the University. If you find yourself in such a situation, it is best to contact your instructor immediately. Link to Academic Integrity Policy: 49-20 Academic Integrity

Disability and Accommodations

If you have accommodations, please be sure to **contact your instructor as soon as possible with details of your accommodation**. Additionally, accommodations letters must be emailed to the Mathematics Department Exam Coordinator, Kyrsten Murphy (kml5346@psu.edu). Any student with a documented disability should contact their instructor as soon as possible so that they can discuss arrangements to fit their needs. For more information, please see the Penn State Educational Equity site: Penn State Educational Equity

Code of Mutual Respect and Cooperation

The Eberly College of Science Code of Mutual Respect and Cooperation pertains to all members of the college community; faculty, staff, and students. The Code of Mutual Respect and Cooperation was developed to embody the values that we hope our faculty, staff, and students possess, consistent with the aspirational goals expressed in the Penn State Principles. The University is strongly committed to freedom of expression, and consequently, the Code does not constitute University or College policy, and is not intended to interfere in any way with an individual's academic or personal freedoms. We hope, however, that individuals will voluntarily endorse the 12 principles set forth in the Code, thereby helping us make the Eberly College of Science a place where every individual feels respected and valued, as well as challenged and rewarded.

Educational Equity

Penn State is a community that fosters both a diverse and inclusive environment for the entire campus community. Acts of intolerance, discrimination, or harassment due to age, ancestry, color, disability, gender, gender identity, national origin, race, religious belief, sexual orientation, or veteran status are not tolerated and can be reported through the Educational Equity on the Report Bias webpage: PSU Report Bias Webpage.

Counseling and Psychological Services

Penn State offers a variety of helpful services on campus that support the mental health of students. Counseling and Psychological Services offers such services as group and individual counseling, psychiatric services as well as virtual chats. For more information you can visit the CAPS website.
Penn State Crisis Line (24 hours/7 days/week): 877-229-6400
Course Coordinators: Nicholas Stepanik and Quinn Minnich
Email: math230coord@psu.edu

Note: The course coordinators reserve the right to modify this syllabus as the academic term progresses.

MATH 230 LEARNING OBJECTIVES

Upon completion of the course, the learner should be able to do each of the following.

Chapter 13: Vectors and 3D Geometry

- Use vectors to describe various mathematical concepts, such as position and direction.
- Perform computations involving vectors including addition, subtraction, multiplication, division, dot product, and cross product.
- Calculate the magnitude/length of a vector.
- Determine the angle between two vectors using appropriate vector computations.
- Normalize a given vector and understand the reasons for using normalized vectors.
- Perform vector projection onto a given vector.
- Perform vector decomposition into parallel and perpendicular vectors to a given vector.
- Calculate and interpret the meaning of the dot product.
- Calculate and, in contextual situations, understand the meaning of cross product.
- Use the cross product to calculate areas of parallelograms and triangles and to calculate volumes of parallelepipeds.
- Find a normal vector to a given pair of vectors, or to a given plane.
- Find the equation of a plane with sufficient given information.
- Sketch planes in $\mathbb{R}^3$.
- Generate graphs of traces of quadric surfaces.
- Identify the type of quadric surface based on traces.
- Use symbolic and graphical techniques to solve problems involving points, vectors, lines, planes, and quadric surfaces in both $\mathbb{R}^2$ and $\mathbb{R}^3$.

Chapter 14: Calculus of Vector-Valued Functions

- Use vector functions to describe concepts of position, velocity, acceleration, among other concepts.
- Parametrize curves in both $\mathbb{R}^2$ and $\mathbb{R}^3$.
- Recognize and understand differences between traditional function notation and parametric form of curves.
- Perform differentiation and integration on vector-valued functions.
- Determine equations of tangent lines using a system of parametric equations.
- Solve initial-value problems for vector-valued functions.
- Use calculus techniques in solving/answering questions concerning position, velocity and acceleration.
- Calculate and interpret the meaning of arc length.
- Determine the arc length parametrization of a curve.

Chapter 15: Calculus of Multivariable Functions

- Evaluate and use multivariable functions.
- Visualize two and three variable functions using graphs, contour maps, and scalar fields.
- Calculate the limit of a multivariable function.
- Understand when a limit exists and does not exist for multivariable functions.
- Calculate approximations of rates of change for multivariable functions.
- Calculate and interpret partial derivatives of multivariable functions.
- Interpret partial derivatives in applications.
- Perform linear approximations of multivariable functions.
- Justify when and how a multivariable function is differentiable both on a given domain and at specified points.
- Determine the equation of a tangent plane to the graph of a function at a given point.
- Calculate the gradient of a multivariable function.
- Understand properties of the gradient and interpret the meaning of the gradient.
- Determine the directional derivative at a given point and provide interpretation.
- Use various multivariable chain rules to calculate rates of change for composite functions.
- Find and classify local extrema of a multivariable function.
- Find and classify global extrema of a multivariable function restricted to a closed and bounded domain.
- Find and classify extrema on a boundary curve or path using either parametrization or Lagrange multipliers.

Chapter 16: Multiple Integration

- Set up and evaluate integrals in $\mathbb{R}^2$ on both rectangular and more general regions.
- Set up and evaluate integrals in $\mathbb{R}^3$ on both boxes and more general regions.
- Apply integration to other contextual examples and problems.
- Calculate average value of a continuous function.
- Use symmetry in evaluating an integral.
- Understand and implement a change in integration order.
- Calculate areas and volumes using multiple integration.
- Convert areas and regions to polar, cylindrical, and spherical coordinates.
- Convert integrals to polar, cylindrical, and spherical coordinates.
- Understand when to use a change of coordinate system for a given situation or integral.
- Convert regions and integrals using a more general transformation.
- Visualize 2D regions under linear and non-linear mapping.
- Calculate the Jacobian of a mapping.
- Convert a given integral to a different coordinate system given a mapping.

Chapter 17: Vector Fields, Vector Calculus, and the Fundamental Theorems

- Understand vector fields in both $\mathbb{R}^2$ and $\mathbb{R}^3$ and their applications.
- Understand the properties of conservative vector fields.
- Determine whether a vector field is conservative using either definitions or theorems.
- Use the Fundamental Theorem of Conservative Vector Fields to calculate the work in a conservative vector field.
- Calculate the curl and divergence of a vector field and understand the interpretation of each.
- Calculate scalar line integrals and their applications.
- Calculate vector line integrals (Work integrals).
- Visually determine whether the net value of a vector line (work) integral is positive, negative, or zero.
- Parametrize a surface in 3D.
- Compute surface integrals using an appropriate surface parametrization.
- Compute the flux of a vector field through a given surface.
- Visually determine the net value of flux as positive, negative, or zero.
- Understand the difference between integrals of scalar function and integrals of lines or surfaces over vector fields.

- Use Green's Theorem to…
  - calculate work along closed curves in $\mathbb{R}^2$.
  - calculate the flux through a closed curve in $\mathbb{R}^2$.
  - calculate area inside of a parametrized closed curve.
- Use Stokes' Theorem to…
  - calculate work along closed curves in $\mathbb{R}^3$.
  - calculate the flux of a solenoidal vector field through surfaces in $\mathbb{R}^3$.
  - understand how surface independence can be used to simplify calculations of work in $\mathbb{R}^3$.
- Use Divergence Theorem to calculate the flux through a closed surface.